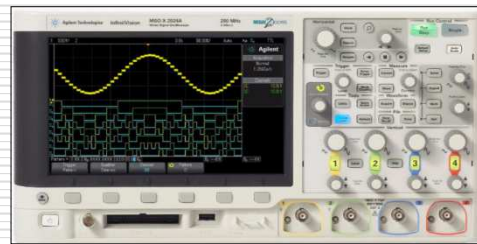


Introduction to laboratory instruments and measurement

(EBN 111 : Electricity and Electronics)

Pre-Practicum 2 Lecture



2013

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Practical 1 Feedback



- ☐ Overall results were good.
- ☐ Was clear that some students did not prepare for the practical.
- ☐ Make sure you use the cursor when reading values from a graph in Orcad.

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Objectives

- ❑ To introduce the student to the basic laboratory instruments and equipment used when building electric circuits.
 - ❑ To familiarize the student with resistance, voltage and current measurement.
 - ❑ To build electric circuits.
- Practical 2 will take place during the week of
22– 26 April

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Two documents

1. Practical 2 Laboratory Guide

- Study this for the practical test

2. Practical 2 Instructions

- Print and bring to the practical session
- Complete during the practical

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Component boxes

- ☐ Contains the components and equipment you need for the practical
- ☐ Connecting wires will be supplied at the practical
- ☐ **Bring your box to the practical**

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Resistors



- ☐ Resistance is specified by a colour code:

Colour	Colour name	1 st stripe (1 st digit)	2 nd stripe (2 nd digit)	3 rd stripe (multiplier)
	Black	0	0	x1
	Brown	1	1	x10
	Red	2	2	x100
	Orange	3	3	x1000
	Yellow	4	4	x10000
	Green	5	5	x100000
	Blue	6	6	x1000000
	Violet	7	7	-
	Grey	8	8	-
	White	9	9	-
	Gold	-	-	x0.1
	Silver	-	-	x0.01

Colour	Colour name	4 th stripe (Tolerance)
	Gold	5%
	Silver	10%

You do not have to memorise these tables.

You may use the leaflet found in your component boxes.

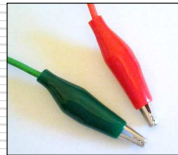
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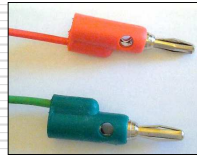
Cables & connectors

- Used to connect your circuit to the lab instruments



Crocodile clips

Connects to:
Your circuit



Banana plugs

Connects to:
Power supplies &
Multimeters



BNC plugs

Connects to:
Oscilloscopes &
Function gen.



HF probe

Connects to:
Your circuit

Not used in EBN111

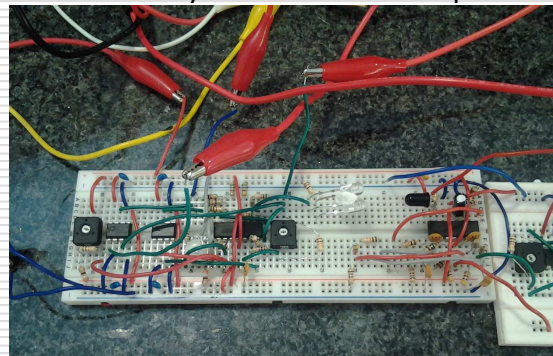
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The protoboard

- Used for building prototype circuits
- You will build all your circuits on a protoboard



Part of a 2-channel infra-red communication system for audio signals.

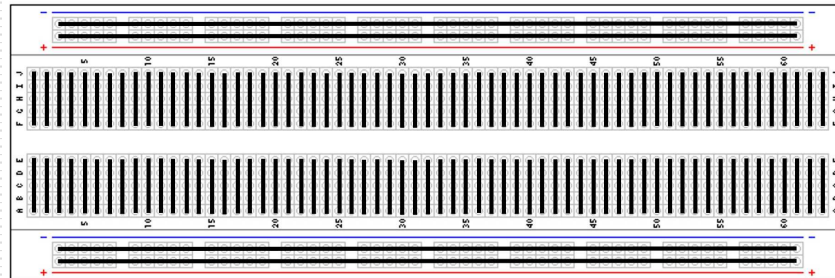
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Protoboard connections

- ☐ Note how the protoboard is connected internally
- ☐ Dark lines are internal metallic connections
- ☐ The long rails are used for power and ground



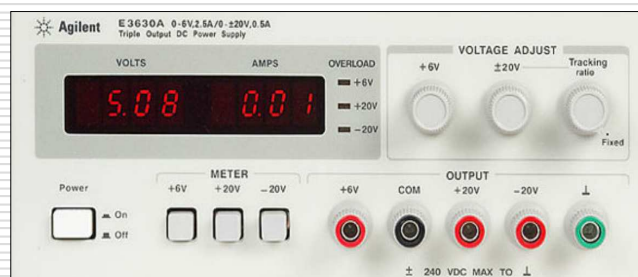
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Laboratory power supply

- ☐ Supplies a variable voltage (voltage source)
- ☐ Limits the current (not a current source)
- ☐ Use COM for ground



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Digital multimeter

- ❑ Measures voltage, current, resistance and more
- ❑ Select the appropriate setting (DCV, DCI or Ω)



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Measurements

- ❑ Voltage measurement
 - Multimeter is connected in PARALLEL with the component
- ❑ Current measurement
 - Multimeter is connected in SERIES with the component
- ❑ Resistance measurement
 - Multimeter is connected in PARALLEL with the component
 - The component/network must be DISCONNECTED FROM THE REST OF THE CIRCUIT

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Oscilloscope and function generator

- ❑ Function generator
 - Generates time varying voltages
 - Allows you to set the waveform, magnitude & frequency
- ❑ Oscilloscope
 - Displays time varying voltages
 - Measures magnitude and frequency

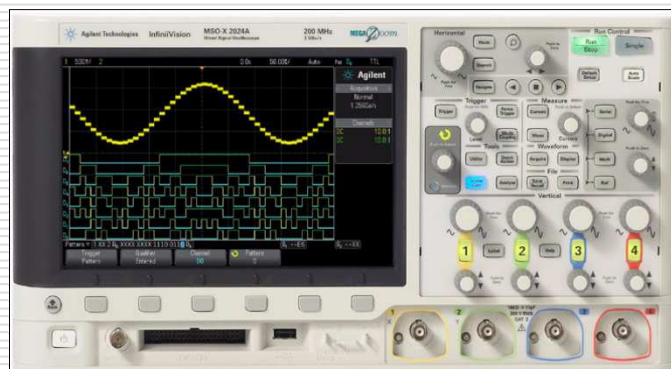
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Oscilloscope and function generator

- ❑ Only the basic functionality will be explored in EBN111



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Preparation

- ☐ **There is NO pre-prac assignment**
- ☐ Familiarise yourself with the contents of your component box
- ☐ Study the Practical 2 Laboratory Guide
 - The practical test will cover the work in this document ONLY.

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Practical Assessment

- ☐ After the test:
 - Complete the Practical 2 Instructions and submit your answers on ClickUp (Practical 2 Online Evaluation)

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Conclusion

- ☐ Take care when using the laboratory instruments
- ☐ The skills learned in this practical are required for practical 3
- ☐ Enjoy it

For questions or feedback
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